Task 3: Linear Regression

Implementation and evaluation of a multiple linear regression model using the Housing dataset.

Objective: To implement and understand linear regression, split data into training and testing sets, train a model, evaluate its performance, and interpret model coefficients.

Dataset: Housing.csv

Tools: Python, Pandas, Matplotlib, and Scikit-learn

Output screenshots and results are taken from the executed Task 3 notebook.

1. Dataset Preview (df.head())

```
price  area  bedrooms  bathrooms  stories mainroad guestroom basement  \
0  13300000  7420         4          2        3      yes        no       no
1  12250000  8960         4          4        4      yes        no       no
2  12250000  9960         3          2        2      yes        no      yes
3  12215000  7500         4          2        2      yes        no      yes
4  11410000  7420         4          1        2      yes       yes      yes

  hotwaterheating airconditioning  parking prefarea furnishingstatus
0              no             yes        2      yes        furnished
1              no             yes        3       no        furnished
2              no              no        2      yes   semi-furnished
3              no             yes        3      yes        furnished
4              no             yes        2       no        furnished
```

2. Dataset Information

```
Dataset Shape: (545, 13)

Column Names:
Index(['price', 'area', 'bedrooms', 'bathrooms', 'stories', 'mainroad',
       'guestroom', 'basement', 'hotwaterheating', 'airconditioning',
       'parking', 'prefarea', 'furnishingstatus'],
      dtype='object')

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 545 entries, 0 to 544
Data columns (total 13 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   price             545 non-null    int64
 1   area              545 non-null    int64
 2   bedrooms          545 non-null    int64
 3   bathrooms         545 non-null    int64
 4   stories           545 non-null    int64
 5   mainroad          545 non-null    object
 6   guestroom         545 non-null    object
 7   basement          545 non-null    object
 8   hotwaterheating   545 non-null    object
 9   airconditioning   545 non-null    object
 10  parking           545 non-null    int64
 11  prefarea          545 non-null    object
 12  furnishingstatus  545 non-null    object
dtypes: int64(6), object(7)
memory usage: 55.5+ KB
```

3. Missing Values and Duplicate Records

```
price             0
area              0
bedrooms          0
bathrooms         0
stories           0
mainroad          0
guestroom         0
basement          0
hotwaterheating   0
airconditioning   0
parking           0
prefarea          0
furnishingstatus  0
dtype: int64
```

4. Categorical Feature Encoding

```
price  area  bedrooms  bathrooms  stories  mainroad  guestroom  \
0  13300000  7420         4          2        3         1          0
1  12250000  8960         4          4        4         1          0
2  12250000  9960         3          2        2         1          0
3  12215000  7500         4          2        2         1          0
4  11410000  7420         4          1        2         1          1
```

```
   basement  hotwaterheating  airconditioning  parking  prefarea  \
0         0                0                1        2         1
1         0                0                1        3         0
2         1                0                0        2         1
3         1                0                1        3         1
4         1                0                1        2         0

   furnishingstatus
0                 0
1                 0
2                 1
3                 0
4                 0
```

5. Input Features and Target Feature

```
Input Features:
   area  bedrooms  bathrooms  stories  mainroad  guestroom  basement  \
0  7420         4          2        3         1          0         0
1  8960         4          4        4         1          0         0
2  9960         3          2        2         1          0         1
3  7500         4          2        2         1          0         1
4  7420         4          1        2         1          1         1

   hotwaterheating  airconditioning  parking  prefarea  furnishingstatus
0                0                1        2         1                 0
1                0                1        3         0                 0
2                0                0        2         1                 1
3                0                1        3         1                 0
4                0                1        2         0                 0

Target Feature:
0    13300000
1    12250000
2    12250000
3    12215000
4    11410000
Name: price, dtype: int64
```

6. Training and Testing Split

```
Training Data: (436, 12)
Testing Data: (109, 12)
```

7. Model Training

```
Linear Regression Model Trained Successfully!
```

8. Sample Predictions

```
Predicted House Prices:
[5203691.70963178 7257004.02115475 3062828.59668171 4559591.65374424
 3332932.30559782 3563080.67918997 5645466.31219972 6413979.66873635
 2755831.54819    2668938.66075228]
```

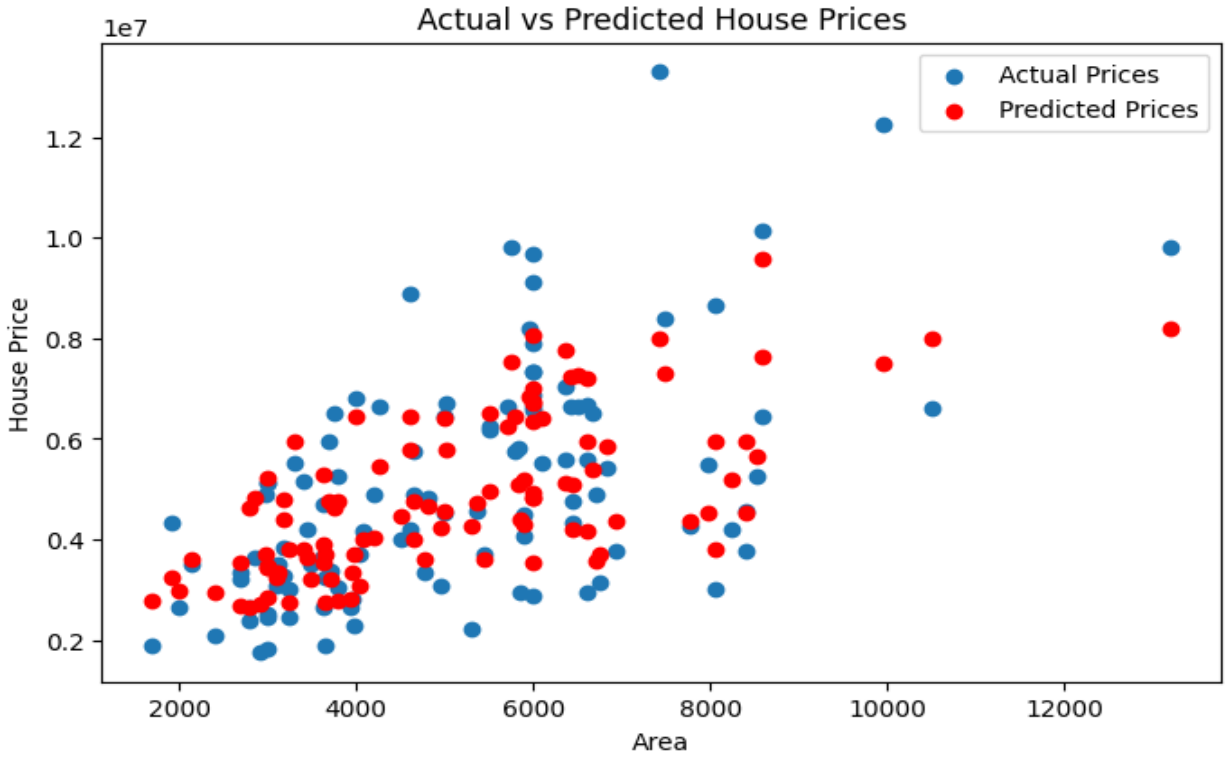
9. Model Evaluation

```
Mean Absolute Error: 979679.6912959901
Mean Squared Error: 1771751116594.0352
R2 Score: 0.6494754192267803
```

10. Actual vs Predicted Values

	Actual Price	Predicted Price
0	4060000	5.203692e+06
1	6650000	7.257004e+06
2	3710000	3.062829e+06
3	6440000	4.559592e+06
4	2800000	3.332932e+06
5	4900000	3.563081e+06
6	5250000	5.645466e+06
7	4543000	6.413980e+06
8	2450000	2.755832e+06
9	3353000	2.668939e+06

11. Actual vs Predicted House Prices Graph



12. Model Intercept and Coefficients

```
Intercept: 293083.06915066857

Coefficients:
area : 235.84877178813724
bedrooms : 78574.49238760071
bathrooms : 1097117.2661479574
stories : 406223.16446288454
mainroad : 366824.1923924831
guestroom : 233146.76562655077
basement : 393159.7787258404
hotwaterheating : 687881.3109570158
airconditioning : 785550.5792954286
parking : 225756.5139762815
prefarea : 629901.6608459124
furnishingstatus : -210397.11827155566
```

13. Coefficient Interpretation

```
area has a positive effect on house price.
bedrooms has a positive effect on house price.
bathrooms has a positive effect on house price.
stories has a positive effect on house price.
mainroad has a positive effect on house price.
guestroom has a positive effect on house price.
basement has a positive effect on house price.
hotwaterheating has a positive effect on house price.
airconditioning has a positive effect on house price.
parking has a positive effect on house price.
prefarea has a positive effect on house price.
furnishingstatus has a negative effect on house price.
```

Conclusion: The model achieved an R^2 score of approximately 0.6495, indicating that it explains about 64.95% of the variation in house prices in the test data. The MAE is approximately 979,680 price units. These results indicate a moderate predictive fit.

Note: Categorical features were label-encoded. Their coefficients should be interpreted with care because numeric labels represent categories, not necessarily equal numerical distances.